

6. Breaking an Affine Cipher

```
#include <stdio.h>

#include <string.h>


int main()
{
    char cipher[200], plain[200];
    int i, c, p;


    printf("Enter ciphertext: ");
    scanf("%s", cipher);


    /* Affine key: a = 3, b = 15
       Inverse of 3 modulo 26 = 9 */


    for(i = 0; cipher[i] != '\0'; i++)
    {
        c = cipher[i] - 'A';

        p = (9 * (c - 15)) % 26;

        if(p < 0)
            p += 26;

        plain[i] = p + 'A';
    }
```

```
plain[i] = '\0';

printf("Decrypted plaintext: %s", plain);

return 0;
}
```

Output

Enter ciphertext: X
Plaintext: U